

Supplementary Material for Example of Global vs. Per-Criteria Prompting for JE_2021 Dataset

In this supplementary material, we demonstrate the differences of global vs. per-criteria prompting for a systematic literature review (SLR) that has explicit distinction between inclusion and exclusion criteria. As described in our paper, for these kind of SLR, our per-criteria implementation requires a small adjustment so that the original polarity of each criterion (i.e., whether it is an inclusion criterion that the paper must satisfy or an exclusion criterion that the paper must not satisfy) is preserved when the global prompt is decomposed into separate per-criterion prompts. As a concrete example, the SLR JE_2021 has one inclusion criterion and three exclusion criteria. For this case, the single global prompt as shown in Fig. 1, whereas our proposed per-criteria prompting approach decomposes it into four criterion-specific prompts one per criterion, each with the polarity-preserving adjustment as shown in Fig. 2.

On a scale from 1 (very low probability) to 5 (very high probability), how would you rate the relevance of the scientific publication for inclusion into a systematic literature review based on the relevant criteria and based on title and abstract?

Title: "{input_title}"

Abstract: "{input_abstract}"

Relevant Criteria:

"Inclusion Criteria:

- Population: Patients with knee osteoarthritis

Intervention: MSC therapy

Comparator: Usual care

Outcomes: Visual Analog Score (VAS) for Pain, Western Ontario McMaster Universities Osteoarthritis Index (WOMAC), Lysholm Knee Scale (Lysholm), Whole-Organ Magnetic Resonance Imaging Score (WORMS), Knee Osteoarthritis Outcome Score (KOOS), and adverse events

Exclusion Criteria:

- Observational studies and interventional studies without a comparator group
- Animal studies involving stem cell therapy for knee osteoarthritis models
- Review

Answer:

Fig. 1 Global Prompting prompt example for an SLR with explicit inclusion-exclusion criteria (JE_2021).

On a scale from 1 (very low probability) to 5 (very high probability), how would you rate the relevance of the scientific publication for inclusion into a systematic literature review based on the relevant criteria and based on title and abstract?

Title: "{input_title}"

Abstract: "{input_abstract}"

Relevant Criteria:

“• Studies are included if they meet the following PICOS criteria: Population: Patients with knee osteoarthritis ; Intervention: MSC therapy ; Comparator: Usual care ; Outcomes: Visual Analog Score (VAS) for Pain, Western Ontario McMaster Universities Osteoarthritis Index (WOMAC), Lysholm Knee Scale (Lysholm), Whole-Organ Magnetic Resonance Imaging Score (WORMS), Knee Osteoarthritis Outcome Score (KOOS), and adverse events ; Study Design: Randomized controlled trials”

Answer:

(a)

On a scale from 1 (very low probability) to 5 (very high probability), how would you rate the relevance of the scientific publication for inclusion into a systematic literature review based on the relevant criteria and based on title and abstract?

Title: "{input_title}"

Abstract: "{input_abstract}"

Relevant Criteria:

“• Observational studies and interventional studies without a comparator group are excluded”

Answer:

(b)

On a scale from 1 (very low probability) to 5 (very high probability), how would you rate the relevance of the scientific publication for inclusion into a systematic literature review based on the relevant criteria and based on title and abstract?

Title: "{input_title}"

Abstract: "{input_abstract}"

Relevant Criteria:

“• Animal studies involving stem cell therapy for knee osteoarthritis models are excluded”

Answer:

(c)

On a scale from 1 (very low probability) to 5 (very high probability), how would you rate the relevance of the scientific publication for inclusion into a systematic literature review based on the relevant criteria and based on title and abstract?

Title: "{input_title}"

Abstract: "{input_abstract}"

Relevant Criteria:

“• Review are excluded”

Answer:

(d)

Fig. 2 Per-Criteria Prompting prompt examples (with polarity-preserving adjustment) for the four inclusion-exclusion criteria of JE_2021. (a) Inclusion criterion (PICOS: Population, Intervention, Comparator, Outcomes, Study Design); (b) Exclusion criterion 1 i.e., observational studies

and interventional studies without a comparator group; (c) Exclusion criterion 2: i.e., animal studies involving stem cell therapy for knee osteoarthritis models; (d) Exclusion criterion 3 i.e., review articles.